
SOFTWARE REQUIREMENTS SPECIFICATION

FOR

WorkNest

VERSION 1.0 APPROVED

PREPARED BY RAHUL SHINDE , ADITYA GHORAD

**Suryadatta Institute of Business Management and Technology
(SIBMT), Pune
Affiliated to Savitribai Phule Pune University (SPPU)**

DATE CREATED:

TABLE OF CONTENTS

TABLE OF CONTENTS.....	II
REVISION HISTORY.....	III
1. INTRODUCTION.....	1
1.1 PURPOSE	1
1.2 DOCUMENT CONVENTIONS	1
1.3 INTENDED AUDIENCE AND READING SUGGESTIONS.....	1
1.4 PRODUCT SCOPE	3
1.5 REFERENCES	4
2. OVERALL DESCRIPTION.....	5
2.1 PRODUCT PERSPECTIVE	1
2.2 PRODUCT FUNCTIONS	1
2.3 USER CLASSES AND CHARACTERISTICS	2
2.4 OPERATING ENVIRONMENT.....	3
2.5 DESIGN AND IMPLEMENTATION CONSTRAINTS.....	4
2.6 USER DOCUMENTATION	6
2.7 ASSUMPTIONS AND DEPENDENCIES	6
3. EXTERNAL INTERFACE REQUIREMENTS.....	1
3.1 USER INTERFACES	1
3.2 HARDWARE INTERFACES	1
3.3 SOFTWARE INTERFACES.....	2
3.4 COMMUNICATIONS INTERFACES	3
4. SYSTEM FEATURES	1
4.1 SYSTEM FEATURE 1	1
4.2 SYSTEM FEATURE 2 (AND SO ON)	1
5. OTHER NONFUNCTIONAL REQUIREMENTS.....	1
5.1 PERFORMANCE REQUIREMENTS.....	1
5.2 SAFETY REQUIREMENTS	1
5.3 SECURITY REQUIREMENTS	2
5.4 SOFTWARE QUALITY ATTRIBUTES	3

5.5 BUSINESS RULES.....	4
6. OTHER REQUIREMENTS.....	6
APPENDIX A: GLOSSARY.....	6
APPENDIX B: ANALYSIS MODELS.....	7
APPENDIX C: TO BE DETERMINED LIST	9

REVISION HISTORY

NAME	DATE	REASON FOR CHANGES	VERSION

1. INTRODUCTION

1.1 PURPOSE

The purpose of this Software Requirements Specification (SRS) document is to formally define the functional and non-functional requirements of **WorkNest**, a web-based role-based workforce management system developed as part of an academic project.

This document serves as a comprehensive reference for understanding system behavior, constraints, and interactions. It provides clear guidance to developers for implementation, to testers for validation, and to faculty for academic evaluation.

WorkNest is designed to provide a **single authentication mechanism** for multiple user roles such as **Super Admin, Admin, Manager, Employee, and Finance**, enabling secure access to a **common dashboard with role-specific features**. The SRS ensures that all stakeholders have a shared understanding of system expectations before development begins.

1.2 DOCUMENT CONVENTIONS

This document follows standard academic and IEEE-style conventions to ensure clarity, consistency, and ease of understanding.

The following conventions are used throughout this Software Requirements Specification (SRS):

- **Requirement Identification:**

- Functional requirements are labeled as **REQ-F-XXX** (e.g., REQ-F-001).
- Non-functional requirements are labeled as **REQ-NF-XXX** (e.g., REQ-NF-001).
- Business rules are labeled as **BR-XX**.
- **Priority Levels:**
 - **High:** Essential requirements mandatory for system operation.
 - **Medium:** Important requirements that enhance usability and performance.
 - **Low:** Optional features and future enhancements.
- **Formatting Standards:**
 - Section headings are numbered hierarchically (e.g., 1, 1.1, 1.1.1).
 - Important terms are highlighted using **bold text**.
 - All requirements are written in clear, concise, and verifiable language.
- **Terminology:**
 - Technical terms such as **JWT**, **RBAC**, **API**, **Dashboard**, and **Authentication** are used consistently and are defined in the Glossary section.
- **Diagrams and Tables:**
 - Diagrams such as ERD, DFD, and sequence diagrams are referenced where applicable.

- Tables are used to present structured information for better readability.

These conventions help maintain uniformity across the document and make it easier for stakeholders to interpret and validate the requirements.

1.3 INTENDED AUDIENCE AND READING SUGGESTIONS

This Software Requirements Specification (SRS) document is intended for the following audiences:

- **Developers:**

Developers will use this document to understand the system requirements, authentication flow, role-based access control, and overall system behavior. Sections 2 (Overall Description) and 4 (System Features) are particularly relevant for implementation.

- **Testers** / **Evaluators:**

Testers and evaluators will refer to this document to verify the system against the specified functional and non-functional requirements. Sections 4 (System Features) and 5 (Nonfunctional Requirements) provide the basis for test case creation and validation.

- **Project Guide** / **Faculty:**

Faculty members and project guides will use this SRS to assess the academic completeness, feasibility, and adherence to software engineering standards. The entire document serves as a reference for evaluation and review.

- **Students** (Project **Owners):**

Students developing the project will use this document as a guideline

throughout the development lifecycle to ensure consistency between requirements, design, and implementation.

- **Reading Suggestions**

- Readers are advised to begin with **Section 1 (Introduction)** to understand the purpose and scope of the system. **Section 2 (Overall Description)** provides a high-level overview of system architecture and users. For detailed system behavior, **Section 4 (System Features)** should be consulted. Quality, performance, and security considerations are covered in **Section 5 (Nonfunctional Requirements)**.
-

1.4 PRODUCT SCOPE

1.5 1.4 Product Scope

WorkNest is a web-based, role-based workforce management system designed to streamline user authentication, access control, and dashboard-based operations within an organization.

The system provides a **single login mechanism** for multiple user roles such as **Super Admin, Admin, Manager, Employee, and Finance**, while ensuring that each role has access only to the features and data relevant to their responsibilities. WorkNest follows a centralized dashboard approach, where the dashboard remains common but its components are dynamically rendered based on the authenticated user's role.

The primary objectives of WorkNest include:

- Secure user authentication and authorization
- Role-based access control to system features
- Protection against unauthorized access and session misuse
- Improved organizational workflow management

WorkNest aims to improve efficiency, security, and clarity in workforce operations by reducing manual processes and enforcing controlled access to system resources.

Note:

The technology stack described in this document represents the **conceptual system design**. Actual implementation technologies may vary while preserving the defined functional behavior and system objectives.

1.5 REFERENCES

The following documents and resources were referred to during the preparation of this Software Requirements Specification (SRS) for the **WorkNest** project:

- **IEEE 830-1998** – *Recommended Practice for Software Requirements Specifications*, IEEE Computer Society.
- **Karl E. Wiegars** – *Software Requirements*, 2nd Edition, Microsoft Press, 2003.
- **Roger S. Pressman** – *Software Engineering: A Practitioner's Approach*, McGraw-Hill Education.

- **MCA Project Guidelines,**
Suryadatta Institute of Business Management and Technology (SIBMT),
Pune,
Affiliated to Savitribai Phule Pune University (SPPU).
 - **Web Technology Documentation**
 - MDN Web Docs – HTML, CSS, JavaScript
 - Node.js Official Documentation
 - React Official Documentation
 - **Internal Project Notes and Design Documents – WorkNest**
(Unpublished).
-

2. OVERALL DESCRIPTION

2.1 PRODUCT PERSPECTIVE

WorkNest is designed as a standalone, web-based workforce management system that follows a modular and layered architecture. The system is developed as an independent application and is not a direct extension or replacement of any existing system.

WorkNest operates on a **client-server architecture**, where the frontend (client) handles user interaction and presentation, while the backend (server) manages business logic, authentication, authorization, and data processing. Communication between the client and server is carried out using secure web protocols.

The system is architected around a **centralized authentication mechanism** and a **role-based access control (RBAC)** model. All users access the system through a single login interface, and after successful authentication, they are redirected to a common dashboard where features and components are displayed dynamically based on the user's assigned role.

From a structural perspective, WorkNest consists of the following major components:

- **Frontend Layer:** Responsible for user interfaces, navigation, and role-based rendering of dashboard components.
- **Backend Layer:** Handles authentication, authorization, business rules, and API services.

- **Data Storage Layer:** Manages persistent storage of user, role, and operational data.

This modular design ensures scalability, maintainability, and flexibility, allowing future enhancements such as reporting modules, analytics, and additional user roles without major architectural changes.

2.2 PRODUCT FUNCTIONS

WorkNest provides a set of core functions that support secure workforce management through role-based access and centralized control. The major functions of the system are summarized below:

- **User Authentication:**
Allows users to log in securely using valid credentials through a single login interface.
- **Role-Based Authorization:**
Determines user access rights based on assigned roles such as Super Admin, Admin, Manager, Employee, and Finance.
- **Centralized Dashboard:**
Provides a common dashboard interface where features and modules are displayed dynamically according to the user's role.
- **User Management:**
Enables authorized roles to create, update, view, and manage user accounts and roles.
- **Task and Workflow Management:**
Allows managers and employees to manage tasks, assignments, and status updates within the system.

- **Access Control and Security:**
Prevents unauthorized access to system resources and ensures protected navigation across the application.
- **Session Management:**
Maintains secure user sessions and ensures proper logout and session termination.

These functions collectively enable WorkNest to improve operational efficiency, enhance security, and provide a structured workflow management environment.

2.3 USER CLASSES AND CHARACTERISTICS

The WorkNest system is designed to support multiple user classes. Each user class has distinct responsibilities, access privileges, and interaction patterns based on organizational roles. The system enforces role-based access control to ensure secure and appropriate usage.

- **Super Admin**
The Super Admin is the highest-level user responsible for system initialization and overall administrative control. This user manages system configurations, creates Admin accounts, and oversees role definitions. The Super Admin is expected to have strong technical knowledge and administrative authority.
- **Admin**
The Admin manages organizational users, roles, and operational data. This role is responsible for employee registration, role assignment, and monitoring system activities. Admin users require moderate technical skills and an understanding of organizational workflows.
- **Project Manager / Manager**
The Project Manager is responsible for creating projects, assigning tasks

to employees, and tracking progress. This user interacts frequently with the system and requires familiarity with task management and reporting features.

- **Employee**

Employees are end users who perform assigned tasks and update their task status. They have restricted access limited to their own tasks, profile, and related information. Basic computer literacy is sufficient for this user class.

Each user class is clearly defined to ensure separation of responsibilities, improved accountability, and enhanced system security.

2.4 OPERATING ENVIRONMENT

The WorkNest system is a web-based application designed to operate in a standard client–server environment. It is accessible through modern web browsers and does not require specialized hardware or software on the client side.

Client Environment

- **Devices:** Desktop computers, laptops, tablets, and smartphones
- **Operating Systems:** Windows, Linux, macOS, Android, iOS
- **Web Browsers:** Google Chrome, Mozilla Firefox, Microsoft Edge, Safari
- **Requirements:** JavaScript enabled, stable internet connection

Server Environment

- **Operating System:** Windows or Linux

- **Backend Runtime:** Node.js (or Java-based backend if required)
- **Database:** MongoDB or equivalent persistent storage system
- **Web Server:** Node.js server / Apache / Nginx (optional for deployment)

Network Environment

- **Internet Connectivity:** Required for accessing the application
- **Protocol:** HTTP/HTTPS
- **Security:** HTTPS recommended for secure communication

This operating environment ensures compatibility, accessibility, and reliable performance across different platforms and devices.

2.5 DESIGN AND IMPLEMENTATION CONSTRAINTS

< The design and implementation of the WorkNest system are subject to the following constraints, which influence the architectural, technological, and operational decisions of the project:

- Technology**

Constraints:

The system is implemented using web-based technologies such as Node.js for the backend and a modern frontend framework. Only widely available and open-source technologies are used to ensure feasibility and cost-effectiveness.
- Security**

Constraints:

User authentication and authorization must follow secure practices. Passwords must be encrypted, and role-based access control must be enforced to prevent unauthorized access to system resources.

- **Architectural**

The system follows a client–server architecture with a clear separation between presentation, application, and data layers. Business logic must not be implemented on the client side.

Constraints:
- **Academic**

The project must comply with academic guidelines, timelines, and evaluation criteria defined by the institute and university.

Constraints:
- **Scalability**

The initial version of the system is designed to support a limited number of users. Advanced scalability and optimization features may be considered in future enhancements.

Constraints:

These constraints ensure that the system remains secure, maintainable, and feasible within the defined project scope.

2.6 USER DOCUMENTATION

The WorkNest system provides essential user documentation to assist users in understanding and effectively using the application. The documentation is designed to be clear, concise, and accessible to users with varying levels of technical expertise.

The following user documentation components are planned for the system:

- **User**

A basic user manual describing system features, login procedures, role-based access, and common workflows such as project and task management.

Manual:

- **Online Help:**
Context-based help and brief instructions provided within the application interface to guide users during operations.
- **System Screens and Tooltips:**
Informative labels and tooltips are used across the user interface to explain buttons, forms, and actions.
- **Error and Notification Messages:**
Clear and user-friendly messages are displayed to inform users about successful operations, errors, or access restrictions.

The user documentation ensures that users can easily learn and operate the system with minimal training and support.

2.7 ASSUMPTIONS AND DEPENDENCIES

The development and operation of the WorkNest system are based on the following assumptions and dependencies. These factors may affect system behavior if they change or are not met.

- **Assumptions**
 - Users will have access to a stable internet connection to use the web-based application.
 - Users will access the system using modern web browsers with JavaScript enabled.
 - Authorized users will follow defined roles and responsibilities while using the system.
 - System administrators will manage user roles and permissions correctly.

- The system will be used within an organizational environment with basic IT infrastructure.

- **Dependencies**

- The system depends on the availability of backend runtime environments such as Node.js or a compatible server platform.
- The application relies on the database or storage system for persistent data management.
- Secure communication depends on proper configuration of HTTPS and related security settings.
- Future enhancements may depend on third-party services or APIs for notifications, analytics, or integrations.

These assumptions and dependencies are considered during system design to ensure reliable and effective operation of the WorkNest system.

3 EXTERNAL INTERFACE REQUIREMENTS

3.1 USER INTERFACES

The WorkNest system provides a clean, responsive, and user-friendly web-based user interface designed to support role-based interaction. The interface is accessible through standard web browsers and adapts to different screen sizes and devices.

- **General Interface Characteristics**

- Web-based graphical user interface (GUI)
- Responsive design for desktop and mobile devices
- Consistent layout and navigation across all pages
- Role-based visibility of menus, buttons, and features
- Clear labels, icons, and action buttons

- **Major User Interface Screens**

- **Login** **Interface:**
Allows users to securely log in using valid credentials. Displays validation messages for incorrect input or unauthorized access.
- **Dashboard** **Interface:**
A common dashboard layout is provided for all users. Dashboard components are dynamically displayed based on the logged-in user's role.

- **User Management Interface:**
Accessible to authorized roles for managing users, roles, and permissions.
- **Project and Task Management Interface:**
Enables project creation, task assignment, progress tracking, and status updates.
- **Profile Interface:**
Displays user-specific information such as role, assigned tasks, and activity history.
- **Interface Usability Features**
 - Input validation and meaningful error messages
 - Confirmation messages for important actions
 - Simple navigation with minimal learning curve

The user interface design focuses on usability, accessibility, and clarity to ensure efficient interaction for all user roles.

3.2 HARDWARE INTERFACES

The WorkNest system is designed as a web-based application and does not require any specialized hardware interfaces. It operates on standard computing devices commonly available in organizational environments.

- **Client-Side Hardware**
 - **Supported Devices:** Desktop computers, laptops, tablets, and smartphones

- **Input Devices:** Keyboard, mouse, and touchscreen (for mobile and tablet devices)
- **Display:** Standard monitor or mobile display with a web browser
- **Server-Side Hardware**
 - **Processor:** Minimum dual-core processor
 - **Memory:** Minimum 2 GB RAM (4 GB recommended)
 - **Storage:** Minimum 20 GB free disk space
 - **Network Interface:** Standard network adapter for internet connectivity

The system interacts with hardware only through standard operating system and browser interfaces. No direct hardware-level communication or specialized peripheral devices are required for the operation of WorkNest.

3.3 SOFTWARE INTERFACES

The WorkNest system interacts with various software components to provide secure and efficient functionality. These interfaces define how different software modules communicate and exchange data.

- **Frontend–Backend Interface**
 - **Frontend Technologies:** Web browser-based interface developed using modern web technologies.
 - **Backend Technology:** Node.js (or compatible server-side framework).
 - **Communication Protocol:** HTTP/HTTPS.
 - **Data Format:** JSON is used for request and response communication.

- **Purpose:** To handle user authentication, role-based authorization, project management, and task operations.
- **Backend–Database Interface**
 - **Database System:** MongoDB or equivalent data storage system.
 - **Access Method:** Backend services interact with the database using secure queries or ORM/ODM mechanisms.
 - **Purpose:** To store and retrieve user data, roles, projects, tasks, and system configurations.
- **Authentication and Security Interfaces**
 - **Authentication Mechanism:** Token-based authentication (JWT).
 - **Authorization:** Role-based access control (RBAC) enforced at the backend.
 - **Password Security:** Encrypted password storage and validation.
- **External Software Dependencies**
 - **Web Browsers:** Google Chrome, Mozilla Firefox, Microsoft Edge, Safari.
 - **Development Tools:** Code editor, runtime environment, and package managers.

These software interfaces ensure seamless interaction between system components while maintaining security, data integrity, and scalability

3.4 COMMUNICATIONS INTERFACES

The WorkNest system uses standard web communication mechanisms to enable secure and reliable data exchange between clients and the server.

- **Network Communication**

- **Communication Protocol:** HTTP/HTTPS
- **Transport Medium:** Internet or local organizational network
- **Client–Server Model:** Requests are initiated by the client (web browser) and responses are handled by the server.

- **Data Transmission**

- **Data Format:** JSON is used for all data exchange between frontend and backend.
- **Request Types:** GET, POST, PUT, DELETE methods are used for communication with backend services.

- **Security and Encryption**

- **Secure Communication:** HTTPS is recommended to encrypt data transmitted between client and server.
- **Session Handling:** Token-based authentication (JWT) is used to maintain secure communication sessions.
- **Access Control:** Unauthorized requests are blocked by backend validation mechanisms.

- **Reliability**

- **Error Handling:** Proper HTTP status codes and messages are used to handle communication errors.
- **Network Dependency:** Stable internet connectivity is required for smooth operation.

These communication interfaces ensure secure, reliable, and efficient interaction between different components of the WorkNest system.

4 SYSTEM FEATURES

This section describes the major functional features of the **WorkNest** system. Each feature represents a core service provided by the system and explains how users interact with it. The system features are defined based on user roles, system behavior, and functional requirements.

The features are organized to clearly describe:

- The purpose of each feature
- The functionality provided by the system
- The interaction between users and the system

Each system feature is further detailed with its description and associated functional requirements. The WorkNest system follows a role-based approach, ensuring that users can access only those features that are permitted according to their assigned roles.

The following subsections describe the individual system features of WorkNest in detail.

4.1 SYSTEM FEATURE 1

4.1.1 Description and Priority

This feature provides secure authentication for all users of the WorkNest system. It allows authorized users to log in using valid credentials and grants access based on assigned roles.

Priority:

High

(Authentication is mandatory for accessing any system functionality.)

4.1.2 Stimulus / Response Sequences

User Action (Stimulus) System Response

User opens login page System displays login form

User enters credentials System validates input

Credentials are valid System authenticates user and redirects to dashboard

Credentials are invalid System displays error message

User logs out System terminates session and redirects to login page

4.1.3 Functional Requirements

- **REQ-F-001:** The system shall allow users to log in using a valid username/email and password.
- **REQ-F-002:** The system shall validate user credentials before granting access.
- **REQ-F-003:** The system shall authenticate users based on their assigned roles.
- **REQ-F-004:** The system shall restrict access to unauthorized users.
- **REQ-F-005:** The system shall redirect authenticated users to a role-based dashboard.
- **REQ-F-006:** The system shall allow users to securely log out and terminate the session.

5 OTHER NONFUNCTIONAL REQUIREMENTS

5.1 PERFORMANCE REQUIREMENTS

The performance requirements define the expected responsiveness and efficiency of the WorkNest system under normal operating conditions.

- **Response** **Time:**
The system shall respond to user actions such as login, dashboard loading, and data submission within **2 seconds** under normal load conditions.
- **Concurrent** **Users:**
The system shall support at least **100 concurrent users** without significant performance degradation.
- **Data Processing Performance:**
Operations such as authentication, role validation, project creation, and task updates shall be processed efficiently to ensure smooth user experience.
- **System** **Availability:**
The system shall be available during standard operational hours with minimal downtime.
- **Scalability:**
The system architecture shall support future performance enhancements and increased user load without major redesign.

These performance requirements ensure that WorkNest delivers a reliable and responsive experience to all users.

5.2 SAFETY REQUIREMENTS

The safety requirements of the WorkNest system focus on preventing accidental data loss, system misuse, and operational failures that could negatively impact users or organizational data.

- Data**

Protection:

The system shall protect user, project, and task data from accidental deletion or corruption.
- Error**

Handling:

The system shall handle system and user errors gracefully by displaying appropriate messages without exposing sensitive information.
- Backup**

and

Recovery:

Regular data backups shall be maintained to prevent data loss in case of system failure.
- Access**

Safety:

Unauthorized users shall be prevented from accessing restricted system features and sensitive data.
- System**

Stability:

The system shall maintain stable operation during normal usage and recover safely from unexpected errors.

These safety requirements ensure that the WorkNest system operates reliably and minimizes risk to users and organizational data.

5.3 SECURITY REQUIREMENTS

The WorkNest system must ensure the confidentiality, integrity, and availability of organizational and user data. Security is a critical requirement due to the role-based nature of the system.

- **Authentication** **Security:**
The system shall authenticate users using secure credentials before granting access to any system feature.
- **Password** **Protection:**
User passwords shall be stored in encrypted or hashed format. Plain-text password storage shall not be permitted.
- **Authorization and Access Control:**
The system shall enforce role-based access control (RBAC) to restrict users from accessing unauthorized features and data.
- **Session** **Management:**
The system shall maintain secure user sessions and invalidate sessions upon logout or prolonged inactivity.
- **Secure** **Communication:**
All data transmitted between the client and server shall use secure communication protocols such as HTTPS.
- **Protection Against Unauthorized Access:**
The system shall prevent unauthorized access attempts and handle security violations appropriately.

These security requirements ensure that WorkNest provides a secure environment for managing organizational operations and sensitive information.

5.4 SOFTWARE QUALITY ATTRIBUTES

The WorkNest system is designed to meet key software quality attributes that ensure usability, reliability, and maintainability of the application.

- **Usability:**

The system shall provide a simple and intuitive user interface that can be easily used by users with basic computer knowledge.

- **Reliability:**

The system shall perform consistently under normal operating conditions and ensure accurate handling of user actions and data.

- **Maintainability:**

The system shall follow a modular design, allowing easy maintenance, debugging, and future enhancements.

- **Scalability:**

The system architecture shall support the addition of new users, roles, and features with minimal changes.

- **Portability:**

The system shall be accessible through standard web browsers across different operating systems.

- **Security:**

The system shall maintain a high level of security to protect sensitive organizational and user data.

These quality attributes ensure that the WorkNest system remains robust, efficient, and adaptable to future needs.

5.5 BUSINESS RULES

The following business rules define the operational policies and constraints that govern the behavior of the WorkNest system. These rules ensure consistency, security, and proper role-based operations within the organization.

- **BR-01:** Only authenticated users shall be allowed to access the WorkNest system.
- **BR-02:** Every user shall be assigned exactly one role at a time, such as Super Admin, Admin, Manager, or Employee.
- **BR-03:** Access to system features and data shall be strictly based on the user's assigned role.
- **BR-04:** Super Admin shall have the highest level of access and authority within the system.
- **BR-05:** Admin users shall be responsible for managing employees, roles, and organizational data.
- **BR-06:** Managers shall be allowed to create projects and assign tasks only to employees under their supervision.
- **BR-07:** Employees shall be permitted to view and update only their assigned tasks and personal profile information.
- **BR-08:** Users shall not be allowed to perform operations outside their authorized permissions.
- **BR-09:** All critical system operations shall be logged for monitoring and accountability purposes.

These business rules ensure structured workflow management, controlled access, and secure operation of the WorkNest system.

6 OTHER REQUIREMENTS

1.6 6. Other Requirements

This section specifies additional requirements of the **WorkNest** system that are not explicitly covered in other sections of this SRS but are necessary for complete system definition.

6.1 Database Requirements

- The system shall use a structured database to store user, role, project, and task information.
- Data consistency and integrity shall be maintained during all create, update, and delete operations.
- Proper indexing and validation mechanisms shall be applied to improve performance and reliability.

6.2 Reusability Requirements

- The system shall be designed using modular components so that modules such as authentication, role management, and task management can be reused in future projects.
- Business logic shall be separated from user interface code to improve reusability.

6.3 Legal and Compliance Requirements

- The system shall ensure basic data privacy by restricting access to authorized users only.
- User data shall be used strictly for organizational and academic purposes.
- The system shall not store or expose sensitive information unnecessarily.

6.4 Internationalization Requirements

- The initial version of the system shall support the English language.
- The system design shall allow future support for multiple languages if required.

6.5 Deployment and Maintenance Requirements

- The system shall be deployable on a standard web server environment.
- Proper documentation shall be maintained to support future maintenance and upgrades.

These additional requirements ensure that the WorkNest system is complete, maintainable, and adaptable to future needs.

APPENDIX A: GLOSSARY

Appendix A: Glossary

The following glossary defines important terms, acronyms, and abbreviations used in this Software Requirements Specification (SRS) for the **WorkNest** system.

Term / Acronym	Description
WorkNest	A web-based role-based workforce management system.
SRS	Software Requirements Specification – a document that defines system requirements.
RBAC	Role-Based Access Control – access permissions based on user roles.
Authentication	Process of verifying a user's identity using credentials.
Authorization	Process of granting access to system features based on role.
User Role	A defined set of permissions assigned to a user (e.g., Admin, Manager).
Dashboard	Central user interface displaying role-specific features.
JWT	JSON Web Token – used for secure user session management.
Session	A period during which a user remains logged into the system.
Frontend	Client-side part of the system responsible for user interaction.
Backend	Server-side part handling business logic and data processing.
API	Application Programming Interface used for communication between systems.
Database	Structured storage for system data such as users and tasks.
Project	A defined set of tasks assigned to employees.
Task	A unit of work assigned to a user within a project.
Admin	User role responsible for managing users and roles.
Super Admin	Highest-level user with full system control.
Employee	User role responsible for executing assigned tasks.

This glossary ensures consistent understanding of terminology across all stakeholders and supports correct interpretation of the SRS document.

APPENDIX B: ANALYSIS MODELS

This appendix describes the analysis models used to understand, visualize, and validate the structure and behavior of the **WorkNest** system. These models help in representing data flow, system interactions, and relationships among system components.

B.1 Entity Relationship Diagram (ERD)

The ERD represents the data structure of the WorkNest system and the relationships between key entities.

Main Entities:

- User
- Role
- Project
- Task

Relationships:

- A **User** is assigned one **Role**
- A **Manager** manages multiple **Projects**
- A **Project** contains multiple **Tasks**
- A **Task** is assigned to one **Employee**

The ERD helps in understanding data storage, integrity, and relationships within the system.

B.2 Data Flow Diagram (DFD)

The Data Flow Diagram illustrates how data flows between users, system processes, and data storage.

Key Processes:

- User Authentication
- Role Validation
- Project Management
- Task Assignment and Tracking

Data Stores:

- User Database
- Project Database
- Task Database

The DFD provides a clear view of system inputs, processing, and outputs.

B.3 Use Case Diagram

The Use Case Diagram represents interactions between user roles and system functionalities.

Actors:

- Super Admin
- Admin
- Manager

- Employee

Major Use Cases:

- Login / Logout
- Manage Users and Roles
- Create Projects
- Assign Tasks
- Update Task Status

This diagram helps in identifying system functionality from the user's perspective.

B.4 Sequence Diagram

The Sequence Diagram illustrates the sequence of interactions between user, frontend, backend, and database.

Example Flow:

- User Login
- Credential Validation
- Role Verification
- Dashboard Access

This model explains the step-by-step flow of system operations.

B.5 State Transition Diagram

The State Transition Diagram shows different states of a user session.

States Include:

- Logged Out
- Logged In
- Active Session
- Session Terminated

This diagram helps in understanding session behavior and transitions.

These analysis models collectively provide a structured and visual representation of the WorkNest system, supporting clear understanding, design validation, and effective implementation.

APPENDIX C: TO BE DETERMINED LIST

The following items in the WorkNest Software Requirements Specification are currently identified as **To Be Determined (TBD)**. These items will be finalized in later stages of design or implementation.

1. **TBD-01:** Final selection of deployment environment (local server or cloud-based hosting).
2. **TBD-02:** Detailed database schema and indexing strategy.
3. **TBD-03:** Session timeout duration and inactivity handling rules.
4. **TBD-04:** Notification and alert mechanisms (email, in-app notifications).
5. **TBD-05:** Advanced reporting and analytics features.

6. **TBD-06:** Integration with third-party services such as email, calendar, or payroll systems.
7. **TBD-07:** Performance optimization strategies for high user load scenarios.

These TBD items will be reviewed and resolved as the WorkNest system progresses through design, development, and testing phases.